

4	(a) (i)	$\tan \theta = \frac{38}{165}$ $\theta = 13^\circ$	C1
		$d \sin \theta = n\lambda$ $d = 2.82 \times 10^{-6} \text{ m}$	C1
		number per metre = $\frac{1}{d} = 3.6 \times 10^5 \text{ m}^{-1}$	A1
	(ii)	Lines further apart in second order, Lines fainter in second order, (if differences stated but without reference to the orders, max 1 mark)	B1
			B1
	(b) (i)	P remains in same position X and Y rotate through 90°	B1
			B1
	(ii)	either screen not parallel to grating or grating not normal to incident light	B1

5	(a) (i)	a region in which a charge will experience a force	B1
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